THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

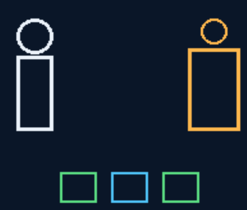
PHASE 238

SIDE-BY-SIDE

THE CREW AND GK3 CAPABILITY SCHOOL

demonstrate, assist, describe

the capability matrix



the chisel is key — the five-driver set — PPE first

VOLTS JOLT · AMPS CRAMP · IMPEDANCE PATH DURATION

CAPABILITY IS DATA, NOT MARKETING

Training & capability — v1.0 in force

GP-IM-238

Revision v1.0 — 24 August 2026

239 of 290

VETTED BATCH 33 — PASS · CAPABILITY TESTING

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CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 238

PHASE 238: SIDE-BY-SIDE — THE CREW AND GK3 CAPABILITY SCHOOL — STATUS: CURRENT — PASS AS A CAPABILITY-TESTING SYSTEM (VET BATCH 33). Crew and GK3 trained side by side, and measured: every skill leaves the school with a mode tag per GK3 unit — DEMONSTRATE (the machine performs it), ASSIST (the machine helps a human do it), DESCRIBE (the machine can only talk a human through it) — the results becoming the standing capability matrix. The distinction the whole phase exists to police: 'the AI understands the instructions' is not 'the machine can physically perform the task'. The curriculum: carpentry (the chisel is key — the tool that punishes bad technique and rewards grain-reading), mechanics (the five-driver fluency set), HVAC (why LPG and natural-gas jets differ, aluminium as sink and reflector, why the air gap works only to 10-20mm, absorption refrigeration, condensation as product and problem), chemistry basics (acids, alkalis, spill discipline), electricity (static versus supply; neutral is sometimes earth, usually not; 'volts give you the jolts, amps give you the cramps' — seated with its third factor: source impedance, current path, duration), power tools and battery discipline (charge discipline is fire doctrine in a closed hull), and the PPE law: almost always gloves — OFF around rotating machinery, because entanglement is worse — and always glasses, the sealed zero-G standard with silicone edge strip, because particles do not fall away up there; they remain on ballistic trajectories until something stops them.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-238, v1.0 — 24 August 2026. The two hundred and thirty-ninth manual of the program — 239 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the school law as seated (the instruction body, the curriculum, the measurement framework) — §2 the framework, graded — §3 the system in the school — §4 the doctrine record — §5 the vet record — §6 standing-up sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status and provenance.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 238: **Side-by-Side — The Crew & GK3 Capability School (Demonstrate/Assist/Describe Mode Tags; the Capability Matrix; Trade Basics — Carpentry, Mechanics, HVAC, Chemistry, Electricity; Volts Jolt, Amps Cramp, Impedance Decides; the Sealed Glasses Standard; Pinball Mechanics Rule the Workplace).** Crew and GK3 train side by side and the school grades the pair: hand tools before power tools, PPE before both, gloves off at rotating machinery, one glasses pattern only.'

THE CORE OBJECTIVE: none seated as a standalone line — the contents line above carries the register wording.

crew staff training along with GK3 side by side training to measure strength agility impact for the GK3. some will be easy others not, we need to know what the can physically teach and help with and what is just verbal and text and pictures.

basic carpentry skills, tennon joints etc, use of a chisel is key,

mechanics, basic screw drivers Allen keys, spanners adjustable wrench, pipe wrench etc

HVAC basics, why gas jets are different sizes in lpg and natural gas , aluminium sink and reflection , why under 100c air gap is the best insulator not insulation, how hot makes cold, condensation as a use and as a byproduct that needs to be managed etc etc

chemistry basics alkaline and acids etc

electricity, static v normal neutral is sometime earth usually not

mostly volage means nothing on the danger scale,

"volts give you the jolts, amps give you the cramps"

power tool usage, battery charging and maintenance.

almost always gloves, always safety glasses, zero g safety glasses standard with silicone edge strip for face seal and elastic head band from the arms, important because the line of fire from an angle grinder in zero g is everywhere, same with woodwork or any particulates use, from the kitchen to the food dryer, basically this is the only type of safety glasses, pinball mechanics rule the workplace.

- THE MEASUREMENT FRAMEWORK (the headline): every skill leaves the school with a mode tag per GK3 unit — DEMONSTRATE (the GK3 physically shows, the crew copies), ASSIST (the GK3 holds, steadies, spots, supplies the strength), or DESCRIBE (verbal, text and pictures only — human hands do the work). Strength, agility and impact are measured, not assumed — the precedent stands: NIST standard test arenas for response robots, the DARPA Robotics Challenge; 'some will be easy others not' is their founding premise too. The output is the CAPABILITY MATRIX, a standing ship document, refreshed whenever a GK3 unit changes. This is the measurement instrument for 225's NP-first law: you cannot spend NP time first until you know what the NP can physically do.

- CARPENTRY: the chisel is key because it is the tool that punishes bad technique and rewards grain-reading — hand skills before power skills, always. The tenon is alignment thinking in wood: the same doctrine as 237's dowel seating. A crew that can cut a tenon can seat a beam.

- MECHANICS: screwdrivers, Allen keys, spanners, adjustable wrench, pipe wrench — the five-driver fluency set; the adjustable and the pipe wrench are the colony's apology tools (they forgive what the kit doesn't cover).

- HVAC — VERIFIED LINE BY LINE (five for five): LPG jets ARE smaller than natural-gas jets (higher calorific value per volume, higher supply pressure — affirmed); aluminium is BOTH reflector (low-emissivity radiant barrier) and heat sink (affirmed); under 100C the still-air gap IS the best insulator ($k \approx 0.026$ W/m·K; bulk insulation is mostly a machine for trapping air) — WITH THE CONVECTION CAVEAT the lesson must carry: size the gap ~10-20 mm or convection cells ferry the heat across; hot-makes-cold IS absorption refrigeration (the gas fridge — no moving parts, colony gold); condensation is BOTH harvest and managed byproduct (235's loop in a teaching coat).

- CHEMISTRY: acids and alkalis, pH, neutralisation, spill discipline — the minimum viable chemistry set for a closed hull.

- ELECTRICITY: static v supply; neutral is SOMETIMES earth, usually not — affirmed and vital (bonded at the source on Earth's MEN systems; floating aboard ship; treat neutral as live, always).

- THE RHYME SEATED WITH ITS THIRD FACTOR: 'volts give you the jolts, amps give you the cramps' — TRUE: ~50-100 mA across the heart is the killer; kilovolts at microamps (static) merely tickle, so voltage alone is mostly not the danger scale. THE COMPLETION (audit insistence): SOURCE IMPEDANCE. Danger = voltage x available current x path through body x duration. A Van de Graaff at 100 kV tickles; a 12 V battery welds a spanner to a hull; 50 V on wet skin can kill. The rhyme stays as spoken; the lesson carries the completion.

- GLOVES — 'ALMOST ALWAYS' AFFIRMED, THE EXCEPTION IS LAW: gloves OFF around rotating machinery — lathes, drill presses, anything that can catch fabric and wrap a hand (degloving is the word the curriculum teaches). The glove rule is per-tool, not per-shift.

- THE SEALED GLASSES STANDARD (the best single design in the phase): zero-g safety glasses with silicone-edge face seal and elastic head band are the ONLY pattern that exists — in zero-g debris never lands: grinder sparks fly straight until they strike, ricochet, and stay aloft; the line of fire is everywhere; PINBALL MECHANICS RULE THE WORKPLACE. From the angle grinder to woodwork to the kitchen to 235's food dryer — one pattern, no second type. ISS crews learned this the hard way with floating swarf; this file learns it on paper.

- POWER TOOLS & BATTERIES: usage, charging, maintenance; charge discipline is fire doctrine in a closed habitat (thermal runaway — the 201/206/207/208 family), NP-monitored charging bay per the NP-first law.
- CURRICULUM ORDER: PPE first, hand tools second, power tools third, trades fourth — and every module closes by measuring BOTH students: the human's competence and the GK3's mode tag. The school grades the pair, never the individual.

INSTRUCTION STATUS (10 August 2026, sitting 147): PARTIAL — the author's voice seated in this body is the surviving fragment of the instruction; the full live order and its thread were not preserved (the 200–260 band was rebuilt by the audit without the chat record). The author counts this phase among the 80 lost instructions. The seated voice is the instruction of record; the audit prose around it is scaffolding.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE MEASUREMENT FRAMEWORK IS THE HEADLINE, AFFIRMED: the phase does not claim the GK3 can magically perform every trade — it sets out to measure what the machine can physically do, per unit, per skill, with the mode tag as the verdict. Capability is data, not marketing.

THE CURRICULUM ORDER IS SAFETY DOCTRINE: PPE first, hand tools second, power tools third, trades fourth — and every module closes by measuring BOTH the crew member and the GK3. The school graduates pairs, not individuals.

THE HVAC MODULE IS VERIFIED LINE BY LINE, FIVE FOR FIVE: LPG jets ARE smaller than natural-gas jets (higher calorific value per volume); aluminium works as radiant reflector and heat sink; still air insulates well but the gap law caps at 10-20mm before convection wakes; absorption refrigeration makes cold from heat; condensation is harvest and hazard both.

THE ELECTRICITY MODULE SEATS THE RHYME WITH ITS THIRD FACTOR: 'volts give you the jolts, amps give you the cramps' — true as mnemonic (roughly 50-100 mA across the heart is the danger band) — and the audit's addition is law: source impedance, current path and duration decide; 'mostly voltage means nothing on the danger scale' survives BECAUSE the path and source are now named. Neutral is sometimes earth, usually not — affirmed and vital on a ship.

THE PPE LAW IS THE BEST SINGLE DESIGN IN THE PHASE: gloves almost always — OFF at rotating machinery, the entanglement exception written as law; glasses always — the sealed zero-G standard, silicone edge strip and elastic, because thrown particles do not fall in free fall: they hold ballistic trajectories until another surface interacts. The glasses are a design answer to a physics fact.

§3 — THE SYSTEM IN THE SCHOOL

- **The mode tags:** DEMONSTRATE / ASSIST / DESCRIBE — per skill, per GK3 unit, into the capability matrix.
- **The pair rule:** every module measures crew and GK3 both — the school graduates pairs.
- **The chisel:** carpentry's key — the tool that punishes bad technique and rewards grain-reading.
- **The five drivers:** screwdrivers, Allen keys, spanners, adjustable, pipe wrench — the fluency set.
- **The rhyme, completed:** volts jolt, amps cramp — and impedance, path, duration decide.
- **Gloves and glasses:** almost always gloves, never at rotation; always glasses, sealed for zero-G.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- The school operationalises 223's latranaut doctrine — the pair structure gains its training arm; the GK3 mediator of 223 is here measured for work, not just argument.
- The capability matrix feeds 233's GK3 install — the deviation-detection algorithm deploys only to units whose matrix says DEMONSTRATE.
- The instruction status is carried honestly: PARTIAL — the author's voice seated in this body is the surviving fragment from sitting 147; the audit framework completes the phase around it.

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT), VET BATCH 33 (17 August 2026): 'Phase 238 — Side-by-Side — PASS as a capability-testing system. This isn't claiming that a GK3 can magically perform every trade. It explicitly sets out to measure what it can physically do: demonstrate; assist; describe. Each capability gets a mode tag, and the result becomes a standing capability matrix. That's exactly the right way to handle the distinction between "the AI understands the instructions" and "the machine can physically perform the task."... The five HVAC teaching points survive the audit... The 10–20 mm air-gap caveat is important... The original "volts jolt, amps cramp" survives as a teaching mnemonic, but the audit correctly adds source impedance, current path and duration. And the glove rule is correctly conditional: gloves off around rotating machinery, because entanglement/degloving is worse... The sealed zero-G glasses rule is also physically coherent: particles don't simply fall away after a tool throws them. 238 → PASS. The remaining work is capability measurement, not fundamental physics.'

§6 — BUILD SEQUENCE

- 1. Write the matrix: skill list, mode-tag definitions, per-unit recording — the standing document.

- 2. Stand up PPE first: the sealed glasses standard issued; the glove rule and its rotating-machinery exception drilled.
- 3. Run hand tools: chisel-first carpentry, the five-driver set — technique measured, not attendance.
- 4. Run the trades modules: HVAC, chemistry, electricity, power tools and batteries — each closed by measuring both pair members.
- 5. Publish the matrix: capability per GK3 unit on the ship's record; the ASSIST and DESCRIBE boundaries explicit.

§7 — CROSS-PHASE (INLINED IN FULL)

- **Phase 223 (the latranaut doctrine):** the pair structure this school trains and measures.
- **Phase 239 (the NPhD mandate):** the matrix is the NPhD's examination — demonstrated capability, tagged.
- **Phase 233 (the planet checker):** the GK3 install draws only from DEMONSTRATE-tagged units.

§8 — DO-NOT-BUILD

- Do NOT tag from specification — the mode tag is measured, per unit; 'understands the instructions' is not a tag.
- Do NOT wear gloves at the lathe — the rotating-machinery exception is law; entanglement is worse.
- Do NOT fly open-frame glasses — zero-G keeps thrown particles ballistic; the sealed standard is the issue.
- Do NOT teach the rhyme bare — impedance, path, duration ride with it, every time.
- Do NOT skip the pair measurement — a module closes only when crew AND GK3 are both measured.

§9 — OWED

OWED: the capability measurement programme itself — the matrix populated per GK3 unit across the curriculum, the mode-tag drift checks (capabilities age with wear and software), and the zero-G glasses' field validation in the particle-throw test.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-238, v1.0 — CURRENT — PASS AS A CAPABILITY-TESTING SYSTEM (VET BATCH 33), seated law, master v2.1 (numerical print order). The two hundred and thirty-ninth manual of the program — 239 of 290. Built 24 August 2026, nineteenth of the 20-manual 'ok ill do these 20, you do 20 more' shift (the author's order, 24 August 2026, seated in sitting 618). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590, 597, 607 and 618): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the 200' / 'ok i will do last nights now you do the next 20' / 'ok ill do these 20, you do 20 more'. The phase's own chain, all seated in the master: the contents line, the bodies (as seated); the live instruction of 10 August 2026 (seated per sitting 147, PARTIAL — the surviving fragment) — the curriculum in his words, including the rhyme verbatim: 'volts give you the jolts, amps give you the cramps'; the demonstrate/assist/describe mode tags and the capability matrix; the five-driver set; the HVAC five-for-five; the gloves-off-at-rotation law and the sealed zero-G glasses standard; the rename of 12 August 2026; the vet batch 33 grade (PASS as a capability-testing system) in §5.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the builders'). Next update triggers: the first fully populated capability matrix, or his word.

END OF MANUAL — PHASE 238